variability in daily production warrants increased maintenance of the older fabricating system? Use the p-value approach.

② New System	① Old System
$\bar{x}_1 = 246$	$\bar{x}_2 = 240$
$s_1 = 15.6$	$s_2 = 28.2$

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 > \sigma_2^2$

$F = \frac{s_1^2}{s_2^2} = \frac{28.2^2}{15.6^2} = 3.268$

p-value $= P(F(29,29) > 3.268)$

p-value $< 0.05 \rightarrow$ reject H_0

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16. SAT Scores The SAT subject tests in chemistry and physics for two groups of 15 students each electing to take these tests are given here.

Chemistry	Physics
$\bar{x} = 784$	$\bar{x} = 758$
$s = 114$	$s = 103$
$n = 15$	$n = 15$

To use the two-sample t-test with a pooled estimate of σ^2, you must assume that the two population variances are equal. Test this assumption using the F-test for equality of variances. What is the approximate p-value for the test?

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$F = \frac{s_1^2}{s_2^2} = \frac{114^2}{103^2} = 1.225$

查 $f_{14,14;0.025} = 2.95$

$1.225 < 2.95 \rightarrow$ non-reject H_0

p-value $= 2 \cdot P(F(14,14) > 1.225)$

$F_{14,14;0.2} = 1.584$

p-value $> 0.4 > 0.05 \rightarrow$ non-reject H_0

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22. Stock Risks The closing prices of two common stocks were recorded for a period of 15 days. The means and variances are

$\bar{x}_1 = 40.33$ $\quad \bar{x}_2 = 42.54$

$s_1^2 = 1.54$ $\quad s_2^2 = 2.96$

a. Do these data present sufficient evidence to indicate a difference between the variabilities of the closing prices of the two stocks for the populations associated with the two samples? Give the p-value for the test and interpret its value.

b. Construct a 99% confidence interval for the ratio of the two population variances.

a. $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$F = \frac{s_2^2}{s_1^2} = \frac{2.96}{1.54} = 1.922$

查 $f_{14,14;0.025} = 2.95$

$1.922 < 2.95 \rightarrow$ Non-reject H_0

p-value $= 2 \cdot P(F(14,14) > 1.922) > 0.2$

b. $\left(\frac{s_1^2}{s_2^2} \cdot \frac{1}{f_{14,14;0.005}}, \frac{s_1^2}{s_2^2} \cdot f_{14,14;0.005}\right) = (0.447, 8.264)$

DATA SET DS1040

11. Reaction Times A comparison of reaction times (in seconds) for two different stimuli in an experiment produced the following results when applied to a random sample of 16 people:

Stimulus 1	1	3	2	1	2	1	3	2
Stimulus 2	4	2	3	3	1	2	3	3

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

$S_1 = 0.83 \quad S_2 = 0.91 \quad \bar{x}_1 = 1.875 \quad \bar{x}_2 = 2.625$

$$S_p^2 = \frac{0.83^2 * 7 + 0.91^2 * 7}{14} = 0.768$$

$$t = \frac{1.875 - 2.625}{\sqrt{S_p^2\left(\frac{1}{7} + \frac{1}{7}\right)}} = -1.713$$

$f_{14, 0.025} = 2.145$

$|-1.713| < 2.145 \rightarrow$ Non-reject H_0

DATA SET DS1041

12. Reaction Times II Refer to Exercise 11. Suppose that the experiment is conducted using eight people as blocks and making a comparison of reaction times within each person; that is, each person is subjected to both stimuli in a random order. The reaction times (in seconds) for the experiment are as follows:

Person	Stimulus 1	Stimulus 2	
1	3	4	-1
2	1	2	-1
3	1	3	-2
4	2	1	1
5	1	2	-1
6	2	3	-1
7	3	3	0
8	2	3	-1

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

$H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$

$\bar{d} = -.75 \quad S_d = 0.8864$

$$t = \frac{-0.75}{\frac{0.8864}{\sqrt{8}}} = -2.393$$

$f_{7, 0.025} = 2.365$

$|-2.393| > 2.365$

$\rightarrow$ reject H_0